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CHITRAKALA

Digital Art Gallery Platform (A Platform to Showcase, Buy, and Sell Local Artworks)

Professionalism Report

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Introduction

I built the Digital Art Gallery Platform because I saw a real problem. Nepali artists especially talented ones from rural areas can't sell their work because they don't have access to physical galleries and can't afford exhibition costs. There are incredible artists doing traditional Thangka, Mithila art, and other forms, but nobody outside their villages ever sees their work.

This report is my honest reflection on what I've built. I'm not going to pretend it's perfect. I've made decisions that could help people, and I've also built in some problems that I need to fix. This is me being clear about both.

1. Social Impact

I. What This Platform Actually Does

Let me be straight about the benefits. An artist in a village outside Kathmandu can now create a profile, upload photos of their work, set their own prices, and sell directly to buyers anywhere in the world. They don't need to pay gallery commission fees. They keep most of the money they earn.

For traditional art forms like Thangka and Mithila art, this is genuinely important. These art forms are being lost because young people aren't learning them there's no money in it. If artists can sell online and actually make a living, that changes things. The platform preserves cultural heritage by documenting it properly with metadata and descriptions.

Buyers also benefit. They get transparent pricing, they can see exactly what they're buying, and they can interact directly with the artist. That's valuable.

II. The Real Problems

But I'd be lying if I didn't acknowledge the complications.

- First, digital access is not equal. Not every artist has reliable internet. Not every artist is comfortable with technology. An elderly artist who does amazing traditional work might not be able to navigate the platform. That's a barrier I can't completely solve in the code it's a social barrier.
- Second, by enabling online sales, I'm potentially reducing reliance on physical galleries and exhibition spaces. That sounds good on the surface, but physical galleries are important culturally and economically. They're part of the art world. If my platform makes people think they don't need galleries anymore, that's a real problem.

- Third, there's the market concentration issue. Popular artists will get more visibility. New artists will struggle to be discovered. The algorithm I've built tries to balance this, but there's no perfect solution. Success in any marketplace is partly luck.

III. Who Wins and Who Doesn't

Honestly, the artists who win most are the ones who already have some advantage maybe they're in Kathmandu, maybe they already have social media presence, maybe they speak English. Rural artists and artists from marginalized communities might struggle more.

For that reason, I'm thinking about adding features that specifically support newer and lesser-known artists. Discoverability tools, mentorship connections, payment plans for buyers so they can afford original pieces. These aren't just nice-to-have features they're part of making the platform actually serve its social purpose.

2. Ethical Considerations

I. The Data Collection Problem

I'm collecting personal data from artists and buyers: names, email addresses, payment information, artwork preferences, browsing history. That's sensitive information. Some of this data could reveal personal things about people someone buying art about social justice, or traditional religious themes, for example.

I didn't implement proper consent mechanisms initially. Users just register and their data gets collected. That's not ethical. They should see clearly what data I'm collecting, why, and for how long.

II. Bias in Recommendations

My recommendation algorithm tries to suggest artworks users might like. But algorithms aren't neutral. If I'm not careful, the system could systematically promote certain artists over others based on factors that aren't fair. Maybe it promotes popular artists more because they have more data. Maybe it unconsciously favors certain art styles.

I've tried to address this by using interpretable logic instead of a black-box model. Users can see why something was recommended to them. But I still need to actively audit the system to check if certain groups of artists are being consistently disadvantaged.

III. Artist Protection and Intellectual Property

Artists upload images of their work. That's valuable intellectual property. I need to prevent people from downloading those images and claiming they're their own work, or using them without permission. My platform doesn't currently have strong protections against this. I should implement watermarking, copyright notices, and report mechanisms.

IV. What I'm Committing To

- First, implementing real consent. When artists or buyers create accounts, they'll see what data is collected and why. They'll be able to control their data.
- Second, auditing the recommendation system. I'll test whether the algorithm is fair across different artist demographics. If certain groups are being systematically disadvantaged, I need to fix that.
- Third, protecting artwork intellectual property. I'll add features that protect artists' work from unauthorized use.

3. Legal Implications

I. Nepal Electronic Transaction Act 2006

Since this platform handles transactions and stores digital records, it's subject to Nepal's Electronic Transaction Act 2006 (<https://dlicca.gov.np/acts/regulation/47>). This law ensures that electronic records are legally valid and that digital transactions have the same legal standing as physical ones.

My platform needs to:

- Maintain integrity of transaction records
- Ensure authenticity of user identities
- Implement secure logging of all transactions
- Follow proper authentication procedures

I'm doing most of this, but I need to ensure transaction logs are properly maintained and can't be altered after the fact.

II. Nepal's Consumer Protection Law

Artists and buyers on my platform are consumers in a marketplace. Nepal's consumer protection framework requires fair dealing and protection against fraud. This means:

1. Clear, honest artwork descriptions (no misleading information)
2. Accurate pricing
3. Fair refund policies
4. Dispute resolution mechanisms

I have these features, but I need to document them more clearly so users understand their rights.

III. If Deployed in the UK or EU: GDPR Compliance

If British or European users access this platform, GDPR applies immediately (<https://gdpr.info.eu/>). This is strict regulation.

Under GDPR, I need:

1. A clear legal basis for processing data (usually "legitimate interest" or "consent")
2. A privacy policy explaining exactly what data is collected and why
3. Mechanisms for users to access their data, correct it, or delete it
4. Data protection impact assessment

Currently, I don't have these implemented properly. Before any EU deployment, I'd need legal review and proper GDPR-compliant systems.

IV. UK Data Protection Act 2018

The UK has its own data protection law (<https://www.legislation.gov.uk/ukpga/2018/12/>). Since I'm developing this at a UK institution, this technically applies. It requires similar protections to GDPR.

V. Copyright and Intellectual Property

Artists retain copyright to their work by default. My platform shouldn't claim ownership of their artwork. The terms of service need to be clear about this. Buyers purchase access to view and potentially own a copy, but the artist maintains intellectual property rights.

4. Security Considerations

I. What I've Done Right

Password storage uses bcrypt hashing, which is industry standard. Passwords are never stored as plaintext. User sessions use JWT tokens with proper expiration.

Database queries use parameterized statements to prevent SQL injection. I'm not building queries by concatenating user input that's the correct approach.

I'm using environment variables for sensitive configuration like API keys. They're not hardcoded in the source code.

II. The Security Problems

But there are issues I need to address.

- **Insufficient Input Validation:** User inputs for artwork descriptions, prices, and comments aren't properly validated. Someone could inject malicious code through their input. I need to sanitize and validate everything.
- **No Rate Limiting:** Someone could spam the system with thousands of login attempts or upload requests. There's nothing stopping brute-force attacks.
- **Insecure File Upload:** Artists upload artwork images, but there's no validation of file types. Someone could upload an executable file disguised as an image. I need to verify file types and scan for malware.
- **Encryption Gaps:** Payment information goes to Stripe, which is secure, but other sensitive data is stored in plaintext in the database. Email addresses, transaction history, user preferences none of this is encrypted.
- **Insufficient Access Control:** The admin functionality doesn't properly verify that only admins can access it. There's no audit trail of who accessed what data.
- **Groq API for Recommendations:** If I ever integrate an external AI service for recommendations, that API call would send user data to external servers. This needs a data processing agreement and should be disclosed to users.

III. Fixes Needed Before Public Launch

Critical:

- Add input validation and sanitization for all user inputs
- Encrypt sensitive data at rest (email, transaction details)
- Implement rate limiting on login and upload endpoints
- Validate and scan uploaded files
- Add proper access control with audit logging

Within one month:

- Implement HTTPS enforcement
- Add security headers to prevent common attacks
- Create a content security policy
- Implement proper error handling that doesn't leak information

Within one/half months:

- Conduct security penetration testing
- Implement database activity monitoring
- Create incident response procedures

5. My Commitments

Building this platform has taught me that code is just part of the job. The harder part is thinking about who could be harmed and building systems that protect them.

I'm committing to:

1. Making consent real. Users will understand what data I'm collecting and have control over it.
2. Actively auditing for bias. I'll test my recommendation system to ensure it's fair across different artist groups.
3. Protecting artist intellectual property. I'll implement features that prevent unauthorized use of their work.
4. Implementing security properly. Not as an afterthought, but from the design phase.
5. Getting legal review before any institutional or public deployment. Someone who understands Nepali and UK law needs to review this.

6. Being honest about limitations. If the platform has accessibility barriers, I'll document that and work to fix it.

This is what being professional means to me. It means taking responsibility for what you build and the people it affects.

6. Closing Reflection

The Digital Art Gallery Platform could do real good. It could help artists make a living from their work and preserve cultural heritage. But it could also create new problems excluding less tech-savvy artists, creating surveillance of user preferences, or harming intellectual property rights if not carefully protected.

I'm proud of what I've built technically. But I'm more committed to building it responsibly. That means fixing the security problems, implementing proper consent, auditing for bias, and thinking about the people who'll use this system.

The responsibility for this platform doesn't end with my code. Artists using it need to understand their rights. Buyers need to be protected. The platform needs to be secure and fair. That takes ongoing commitment from me and from anyone who deploys this system.

6. References

1. **Nepal Electronic Transaction Act, 2006**

https://radiantca.com.np/assets/nav_file/Electronic%20Transaction%20Act%202063.pdf

2. **GDPR - General Data Protection Regulation (EU) 2016/679**

<https://gdpr-info.eu/>

3. **UK Data Protection Act 2018**

<https://www.legislation.gov.uk/ukpga/2018/12/contents>